



# **Predictive Fault Detection in Robots Using Deep Learning**

## **Project Proposal**

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## Abstract

Robots are widely used in industries, autonomous vehicles, and space exploration. One big challenge is that robots can fail unexpectedly due to problems in their motors, battery, or other internal systems. Most existing methods detect these faults only after they happen, which can cause delays, damages, or accidents. This project aims to predict robot failures **before** they happen using deep learning. A simulated robot will operate in a simple environment, and its internal data—such as speed, acceleration, battery level, motor load, and task time—will be recorded. Using this historical data, a deep learning model called **LSTM** (Long Short-Term Memory) or **GRU** (Gated Recurrent Unit) will learn patterns that happen before a failure occurs. The model will output a fault risk percentage for the robot at each step. For example, it can warn: “Battery low, motor load high, predicted risk 80%, reduce speed or stop robot.” This approach allows the robot to take action **before a failure happens**, making it more reliable. This project is fully robotics-related because it focuses on robot health and predictive maintenance, not just vision or navigation. It combines robotics simulation with deep learning and gives clear, measurable results that can be tested in Python.

## 1. Introduction

Robots are increasingly used in industries, autonomous vehicles, and automated systems to carry out important tasks. Despite their design for reliability, robots can still experience unexpected failures due to issues such as motor overload, low battery, or other internal system problems. Traditional fault detection methods usually identify these problems **after** they occur, which can lead to downtime, equipment damage, safety hazards, and higher costs. This reactive approach is inefficient, especially in critical robotics applications. To overcome this, the project focuses on **predictive fault detection**, which aims to anticipate failures before they happen. A simulated robot will operate in a Python-based environment, performing tasks while tracking internal parameters such as speed, acceleration, battery level, motor load, and task duration. A deep learning model, like LSTM or GRU, will be trained on this historical data to identify patterns that indicate an imminent failure. The model will provide a risk score at each step, enabling the system to suggest preventive actions such as slowing down or stopping the robot. This proactive method enhances robot reliability, safety, and efficiency, making it highly relevant to real-world robotics applications where preventing failures is crucial.

## 2. Goals and Objectives

The Goals and Objectives are as follow:

1. To develop a system that can predict robot failures before they occur, improving reliability and safety.
2. To create a simple Python-based robot simulation that logs internal parameters such as speed, acceleration, battery level, motor load, and task duration.
3. To identify patterns in the robot’s behavior that indicate an imminent fault.
4. To train a deep learning model (LSTM or GRU) on historical robot data for predictive fault detection.
5. To provide a fault risk score and actionable recommendations, such as slowing down or stopping the robot, at each step of operation.
6. To evaluate the performance of the predictive system using metrics like accuracy, precision, and recall.

7. To demonstrate the practical applicability of predictive fault detection in simulated robotics environments for proactive maintenance.

### 3. Scope of the Project

This project focuses on predicting robot failures before they occur using a simulated environment. The scope includes creating a Python-based robot simulator that records internal parameters such as speed, acceleration, battery level, motor load, and task duration. A deep learning model (LSTM or GRU) will be trained on this data to identify patterns that indicate an imminent fault. The system will provide a fault risk score and recommendations at each step, helping prevent failures proactively. The project does not include real hardware implementation or external sensors; it is limited to simulation and predictive analysis. The results will demonstrate how predictive fault detection can improve robot reliability, safety, and efficiency, making it relevant to industrial, autonomous, and space robotics applications.

### 4. Initial Study and Work Done so Far

For this project, we studied how robots can fail unexpectedly and how predictive maintenance can help prevent such failures. Most existing methods detect faults only after they occur, so we aim to build a system that can **predict problems before they happen** using deep learning models like LSTM or GRU. We are using publicly available predictive maintenance datasets that contain time-series data of machine or robot parameters and failure events. From these datasets, we analyze important parameters like speed, acceleration, battery level, motor load, and task duration to understand patterns that lead to failures. At the same time, we have started designing a **Python-based robot simulator**, where a virtual robot performs tasks and its internal parameters are recorded. This simulation will allow us to test the deep learning model in a controlled environment. So far, we have focused on understanding the datasets, identifying key parameters that affect robot health, and planning the model and simulation setup to achieve accurate predictive fault detection.

### References

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- [2] Forecasting failure rate of IoT devices: A deep learning way to predictive maintenance <https://dl.acm.org/doi/10.1016/j.compeleceng.2023.108829>
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